AGRIVIVE TECHNICAL SPECIFICATION

AHP-TOPSIS Ranking Engine & n8n Marketing Boost Automation

Mathematical Formulations, Multi-Criteria Synthesis, and Headless Social Promotion Architecture

Project: Agrivive Surplus Vegetable Marketplace | Location: Davao City, Philippines
Authors: Bartiquien, Jara, Marundan | Date: November 8, 2026 | Version: 2.5.0

EXECUTIVE SUMMARY & SPECIFICATION SCOPE

This specification presents the complete algorithmic and architectural blueprints for Agrivive's core visibility and marketing systems: (1) The AHP-TOPSIS Multi-Criteria Decision Visibility Engine, which ranks surplus vegetable listings dynamically and fairly without pay-to-win mechanics, paired with the 6-Hour Boost Rotation Rule to guarantee equitable exposure; and (2) The n8n Marketing Boost and Dynamic Price-Drop Engine, a decoupled webhook-driven automation system utilizing Groq AI copywriting and Meta Graph API syndication to convert aging stall surplus into viral community sales before end of day.

Document Pipeline Structure: In accordance with system specifications, each core component is presented following a strict progression: Visual Pipeline Diagram → System Description & Grounding → Step-by-Step Mathematical/Payload Calculations. Every calculation is isolated into a numbered table containing raw or transformed data, formal mathematical expressions, and exhaustive explanatory context.

PART 1: AHP-TOPSIS VISIBILITY RANKING PIPELINE

1.1 Pipeline Architecture & Execution Flow

The AHP-TOPSIS visibility engine operates on four raw produce attributes extracted directly from PostgreSQL. The pipeline synthesizes subjective multi-stakeholder priorities via Thomas Saaty's Analytic Hierarchy Process (AHP) and executes multi-dimensional geometric distance optimization via Hwang & Yoon's Technique for Order Preference by Similarity to Ideal Solution (TOPSIS). The terminal stage applies Agrivive's 6-Hour Lock Rule to determine automated boost eligibility.

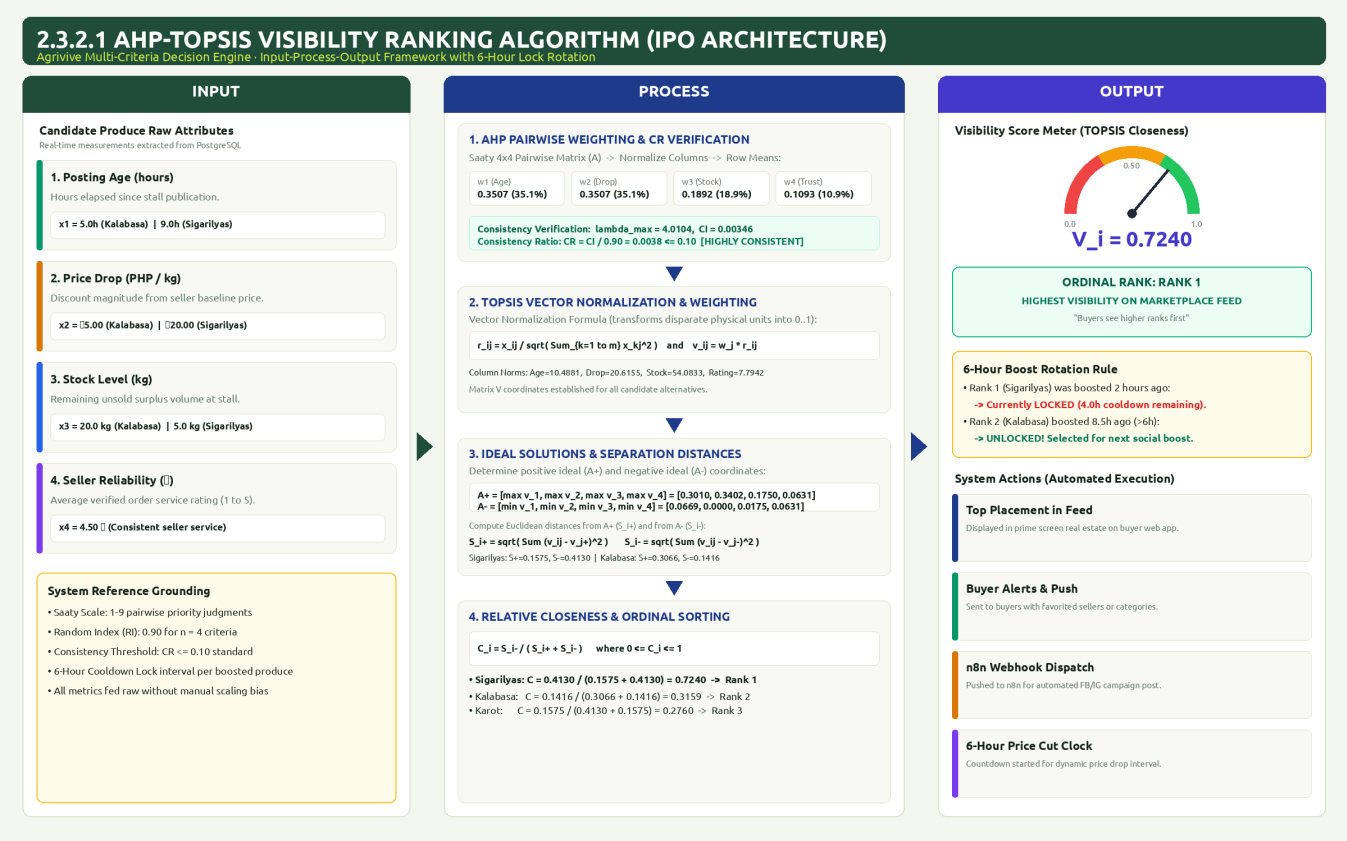


Figure 1: AHP-TOPSIS Input-Process-Output (IPO) Architecture with 6-Hour Lock Rotation Framework.

1.2 System Description & Theoretical Grounding

Agricultural surplus marketplaces face unique inventory dynamics: perishable fresh produce loses nutritional and market value rapidly over hours. Traditional e-commerce algorithms favor paid placements or historical cumulative sales, which systematically penalize smallholder farmers and newly listed fresh surplus. Agrivive eliminates 'pay-to-win' bias through multi-criteria decision modeling (MCDM):

- C_1: Posting Age (Freshness Urgency): Elapsed hours since listing cycle publication. Older surplus must be highlighted before spoilage, maximizing waste diversion.
- C_2: Price Drop (Discount Magnitude): Price per unit reduced from the seller's original baseline (Base Price − Current Price). Directly incentivizes dynamic discounts.
- C_3: Stock Level (Surplus Volume): Remaining unsold inventory in kilograms. Higher volume listings require greater visibility to clear stall tables before dusk.
- C_4: Seller Reliability (Service Trust): Average service star rating (1.0 to 5.0) from verified completed orders. Protects buyers from unfulfilled reservations.

Removal of Tiers in Favor of Pure Ordinal Ranking: In early iterations, produce was categorized into static tiers ('Priority Boost', 'Standard Exposure', 'Basic'). This caused boundary clipping issues where listings clustered near score thresholds (e.g. 0.59 vs 0.60) experienced jarring visibility drops. Agrivive now utilizes pure Ordinal Ranking (Rank 1, Rank 2, Rank 3, ... Rank N). Boost rotation is governed objectively by the 6-Hour Lock Rule: once boosted, a vegetable enters a 6-hour cooldown lock, allowing the next-ranked unlocked produce to receive automated promotion.

1.3 Step-by-Step Mathematical Calculations

The following tables demonstrate the complete end-to-end execution of the AHP-TOPSIS engine using real candidate produce data from Agrivive's test harness (squash [Kalabasa], carrots [Karot], and winged beans [Sigarilyas]).

Table 1: Candidate Produce Raw Evaluation Matrix (X_raw)

MATHEMATICAL EXPRESSION & FORMULATION:

$$x_2 = \max(0, \text{Price}_{base} - \text{Price}_{current})$$

$$x_1 = \max(0, (t_{now} - t_{published}) / 3600)$$

Alternative Produce	Posting Age (x_1)	Base Price	Current Price	Price Drop (x_2)	Stock Level (x_3)	Seller Reliability (x_4)
A_1: Kalabasa (Squash)	5.0 hours	₱50.00 / kg	₱45.00 / kg	₱5.00 / kg	20.0 kg	4.50 / 5.0 (verified)
A_2: Karot (Carrot)	2.0 hours	₱60.00 / kg	₱60.00 / kg	₱0.00 / kg	50.0 kg	4.50 / 5.0 (verified)
A_3: Sigarilyas (Winged Bean)	9.0 hours	₱80.00 / kg	₱60.00 / kg	₱20.00 / kg	5.0 kg	4.50 / 5.0 (verified)

Explanation & Interpretation: Captures the initial unscaled state of active candidate vegetables at a Davao City market stall. Sigarilyas has been on display the longest (9h) and features the largest price reduction (₱20/kg) but lowest remaining stock (5kg). Karot is fresh (2h) with large inventory (50kg) but zero price reduction. Kalabasa displays moderate age and discount. All three alternatives currently share an identical high seller reliability rating (4.50 stars).

Table 2: AHP Pairwise Comparison Matrix (A)

MATHEMATICAL EXPRESSION & FORMULATION:

$$A = [a_{ij}]_{4\times4} \text{ such that } a_{ij} \in \{1, 2, \ldots, 9\}, a_{ji} = 1 / a_{ij}, a_{ii} = 1.0$$

Criteria	C_1: Posting Age	C_2: Price Drop	C_3: Stock Level	C_4: Reliability
C_1: Posting Age (Freshness)	1.0000	1.0000	2.0000	3.0000
C_2: Price Drop (Discount)	1.0000	1.0000	2.0000	3.0000
C_3: Stock Level (Volume)	0.5000 (1/2)	0.5000 (1/2)	1.0000	2.0000
C_4: Seller Reliability	0.3333 (1/3)	0.3333 (1/3)	0.5000 (1/2)	1.0000

Explanation & Interpretation: Establishes relative decision weights based on domain requirements for surplus recovery. Posting age (C_1) and price discount (C_2) are judged equally critical (1.0). Each is twice as important as stock volume (2.0) and three times as important as seller reliability (3.0) because surplus perishability and price sensitivity drive rapid pickup.

Table 3: AHP Column Summation Vector (S_col)

MATHEMATICAL EXPRESSION & FORMULATION:

$$S_j = \sum_{k=1}^{n} a_{kj} \text{ for each criterion column } j \in \{1, 2, 3, 4\}$$

Row Vector Component	C_1: Posting Age	C_2: Price Drop	C_3: Stock Level	C_4: Reliability
Sum of Column Values: S_j	2.8333 (17/6)	2.8333 (17/6)	5.5000 (11/2)	9.0000 (9/1)

Explanation & Interpretation: Computes the aggregate magnitude of each column in matrix A. These sums serve as normalization denominators in the next step to transform pairwise ratios into proportional fractions totaling 1.0 per column.

Table 4: Normalized AHP Comparison Matrix (A_norm)

MATHEMATICAL EXPRESSION & FORMULATION:

$$\bar{a}_{ij} = a_{ij} / S_j = a_{ij} / \left(\sum_{k=1}^{n} a_{kj} \right)$$

Criteria	C_1: Age	C_2: Drop	C_3: Stock	C_4: Reliability
C_1: Posting Age	0.3529	0.3529	0.3636	0.3333
C_2: Price Drop	0.3529	0.3529	0.3636	0.3333
C_3: Stock Level	0.1765	0.1765	0.1818	0.2222
C_4: Seller Reliability	0.1176	0.1176	0.0909	0.1111

Explanation & Interpretation: Divides each matrix entry by its corresponding column sum. Each column now sums exactly to 1.0000, enabling unbiased averaging across rows to extract criteria weights.

Table 5: AHP Criteria Priority Weight Vector (w)

MATHEMATICAL EXPRESSION & FORMULATION:

$$w_i = (1 / n) \sum_{j=1}^{n} \bar{a}_{ij}, \text{ where } n = 4 \text{ criteria and } \sum_{i=1}^{n} w_i = 1.0000$$

Criterion Identifier	Criterion Description	Priority Weight (w_i)	Weight Percentage (%)
w_1 (Posting Age)	Urgency of fresh harvest clearance	0.3507	35.07%
w_2 (Price Drop)	Consumer economic discount incentive	0.3507	35.07%
w_3 (Stock Level)	Surplus inventory clearance priority	0.1892	18.92%
w_4 (Seller Reliability)	Historical pickup and fulfillment trust	0.1093	10.93%
Sum / Validation	Total criteria weight budget	1.0000	100.00%

Explanation & Interpretation: Calculated as the arithmetic mean of each row in A_norm. The resulting distribution establishes that posting age (35.07%) and price discounts (35.07%) govern over 70% of marketplace placement, with inventory volume (18.92%) and seller service trust (10.93%) forming the remainder.

Table 6: AHP Mathematical Consistency Verification

MATHEMATICAL EXPRESSION & FORMULATION:

$$CI = (\lambda_{max} - n) / (n - 1) = 0.00346$$

$$CR = CI / RI_4 = 0.00346 / 0.90 = 0.00384 \leq 0.10 \text{ (Statistically Consistent)}$$

Metric Name	Mathematical Expression	Calculated Value	Threshold & Decision
Weighted Sum Vector (WSV)	WSV = A · w	[1.4079, 1.4079, 0.7587, 0.4377]	Intermediate Vector
Consistency Vector (CV)	CV_i = WSV_i / w_i	[4.0143, 4.0143, 4.0091, 4.0036]	Approaches n = 4
Max Eigenvalue (λ_max)	λ_max = (1/n) ∑ CV_i	4.0104	λ_max ≥ n (4.0)
Consistency Index (CI)	CI = (λ_max − n) / (n − 1)	0.00346	Low dispersion
Random Index (RI)	Empirical RI for n = 4	0.9000	Saaty reference table
Consistency Ratio (CR)	CR = CI / RI	0.00384 (0.38%)	PASSED (CR ≤ 0.10)

Explanation & Interpretation: AHP requires verifying that human judgment transitivities do not contain severe contradictions. The calculated Consistency Ratio of 0.00384 (0.38%) is vastly below the maximum allowable threshold of 0.10 (10.0%), proving that the extracted weights are logically consistent and formally approved for TOPSIS ranking.

Table 7: TOPSIS Euclidean Vector Normalization Denominators

MATHEMATICAL EXPRESSION & FORMULATION:

$$||x_{*_j}|| = \sqrt{\sum_{i=1}^{m} x_{ij}^2} \text{ for each criterion column } j \text{ across } m = 3 \text{ alternatives}$$

Vector Parameter	C_1: Age (hours)	C_2: Drop (PHP)	C_3: Stock (kg)	C_4: Reliability (Score)
Sum of Squared Values: $\sum x_{ij}^2$	$5^2+2^2+9^2 = 110.00$	$5^2+0^2+20^2 = 425.00$	$20^2+50^2+5^2 = 2925.00$	$4.5^2 \times 3 = 60.75$
Euclidean Norm: $\sqrt{(\sum x_{ij}^2)}$	10.4881	20.6155	54.0833	7.7942

Explanation & Interpretation: Calculates the quadratic root sum of squares for each column in the decision matrix X. These divisors normalize disparate physical units (hours, pesos, kilograms, star ratings) into standardized scale lengths.

Table 8: TOPSIS Normalized Decision Matrix (R)

MATHEMATICAL EXPRESSION & FORMULATION:

$$r_{ij} = x_{ij} / \sqrt{\sum_{k=1}^{m} x_{kj}^2}$$

Alternative	C_1: Age (r_i1)	C_2: Drop (r_i2)	C_3: Stock (r_i3)	C_4: Reliability (r_i4)
A_1: Kalabasa	5 / 10.4881 = 0.4767	5 / 20.6155 = 0.2425	20 / 54.0833 = 0.3698	4.5 / 7.7942 = 0.5774
A_2: Karot	2 / 10.4881 = 0.1907	0 / 20.6155 = 0.0000	50 / 54.0833 = 0.9245	4.5 / 7.7942 = 0.5774
A_3: Sigarilyas	9 / 10.4881 = 0.8581	20 / 20.6155 = 0.9701	5 / 54.0833 = 0.0925	4.5 / 7.7942 = 0.5774

Explanation & Interpretation: Matrix R converts all raw measurements into dimensionless values bounded between 0 and 1. Sigarilyas dominates the normalized posting age (0.8581) and price drop (0.9701) columns, while Karot dominates inventory volume (0.9245).

Table 9: TOPSIS Weighted Normalized Decision Matrix (V)

MATHEMATICAL EXPRESSION & FORMULATION:

$$v_{ij} = w_j \cdot r_{ij} \text{ where } w_j \text{ is the derived AHP weight vector}$$

Alternative	C_1: Age (v_i1)	C_2: Drop (v_i2)	C_3: Stock (v_i3)	C_4: Reliability (v_i4)
Applied Weight w_j	$w_1 = 0.3507$	$w_2 = 0.3507$	$w_3 = 0.1892$	$w_4 = 0.1093$
A_1: Kalabasa	0.1672	0.0851	0.0700	0.0631
A_2: Karot	0.0669	0.0000	0.1750	0.0631
A_3: Sigarilyas	0.3010	0.3402	0.0175	0.0631

Explanation & Interpretation: Applies the AHP criteria priorities to the normalized matrix. This weights the geometric hyperspace, ensuring that high scores in Age and Price Drop carry 3.2 times the coordinate distance impact of Seller Reliability.

Table 10: Positive Ideal (A^+) and Negative Ideal (A^-) Solutions

MATHEMATICAL EXPRESSION & FORMULATION:

$$A^+ = [\max_i(v_{ij})] = [0.3010, 0.3402, 0.1750, 0.0631]$$
$$A^- = [\min_i(v_{ij})] = [0.0669, 0.0000, 0.0175, 0.0631]$$

Reference Boundary Vector	C_1: Age (v_1)	C_2: Drop (v_2)	C_3: Stock (v_3)	C_4: Reliability (v_4)
Positive Ideal Solution (A^+)	max = 0.3010	max = 0.3402	max = 0.1750	max = 0.0631
Negative Ideal Solution (A^-)	min = 0.0669	min = 0.0000	min = 0.0175	min = 0.0631

Explanation & Interpretation: Identifies the theoretical best and worst composite listings across the active marketplace cohort. Because all four criteria are treated as benefit attributes ('higher is better for clearance and trust'), A^+ selects column maximums and A^- selects column minimums.

Table 11: TOPSIS Euclidean Distance Separation Vectors

MATHEMATICAL EXPRESSION & FORMULATION:

$$S_i^+ = \sqrt{\left(\sum_{j=1}^{n} (v_{ij} - v_j^+)^2 \right)}$$
$$S_i^- = \sqrt{\left(\sum_{j=1}^{n} (v_{ij} - v_j^-)^2 \right)}$$

Alternative Produce	Distance to Best (S_i^+)	Distance to Worst (S_i^-)	Sum of Distances ($S^+ + S^-$)
A_1: Kalabasa	0.3066	0.1416	0.4483
A_2: Karot	0.4130	0.1575	0.5704
A_3: Sigarilyas	0.1575	0.4130	0.5704

Explanation & Interpretation: Calculates multi-dimensional Euclidean geometric distances. Sigarilyas is extremely close to the ideal best ($S^+ = 0.1575$) and far from the worst ($S^- = 0.4130$). In contrast, Karot is farthest from the ideal best ($S^+ = 0.4130$).

Table 12: Final Relative Closeness Coefficient & Ordinal Ranking

MATHEMATICAL EXPRESSION & FORMULATION:

$$C_i = S_i^- / (S_i^+ + S_i^-) \text{ where } 0 \le C_i \le 1$$
$$\text{Ordinal Rank} = \text{argsort}(C_i, \text{descending} = \text{True})$$

Alternative Produce	Closeness Ratio Expression	Relative Closeness (C_i)	Ordinal Rank	Placement Effect
A_3: Sigarilyas	0.4130 / (0.1575 + 0.4130)	0.7240	**Rank 1**	Top of Marketplace Feed
A_1: Kalabasa	0.1416 / (0.3066 + 0.1416)	0.3159	**Rank 2**	Middle of Marketplace Feed
A_2: Karot	0.1575 / (0.4130 + 0.1575)	0.2760	**Rank 3**	Lower Section of Feed

Explanation & Interpretation: Computes the final TOPSIS score C_i. Higher values indicate greater proximity to A^+ and separation from A^-. Sigarilyas achieves Rank 1 (0.7240), Kalabasa achieves Rank 2 (0.3159), and Karot achieves Rank 3 (0.2760). The engine outputs clean ordinal ranks for consumer display ('Buyers see higher ranks first').

Table 13: Agrivive 6-Hour Lock Rule & Automated Boost Selection

MATHEMATICAL EXPRESSION & FORMULATION:

$$t_{locked_until} = t_{last_boost} + 6 \text{ hours}$$
$$is_locked = (t_{now} < t_{locked_until})$$
$$\text{Boost Target} = \text{argmin}_{rank}(\text{Candidate} \mid is_locked = \text{False})$$

Produce	Rank	Last Boosted Timestamp	Elapsed Time	Cooldown Status	Boost Decision & Action
A_3: Sigarilyas	Rank 1	2.0 hours ago	2.0 h < 6.0 h	LOCKED (until +4.0h)	SKIPPED (Active lock cooldown)
A_1: Kalabasa	Rank 2	8.5 hours ago	8.5 h ≥ 6.0 h	UNLOCKED (Eligible)	SELECTED FOR BOOST (n8n Webhook Sent)
A_2: Karot	Rank 3	Never boosted	N/A	UNLOCKED (Eligible)	STANDBY (Lower rank than Kalabasa)

Explanation & Interpretation: Demonstrates Agrivive's anti-monopoly boost rotation. Even though Sigarilyas ranks #1, it was boosted 2 hours ago and remains locked. The background worker traverses down the ordinal ranks to Rank 2 (Kalabasa), which has exceeded the 6-hour lock threshold. Kalabasa is selected, and its webhook payload is dispatched to n8n.

PART 2: N8N MARKETING BOOST & SOCIAL AUTOMATION ENGINE

2.1 Pipeline Architecture & Execution Flow

The Agrivive Boost pipeline automates social media syndication and scheduled price drops through a headless n8n workflow. It features two asynchronous execution pathways triggered by a single secure webhook endpoint: Branch A (Initial Boost) creates authentic promotional captions and posts them to Meta (Facebook & Instagram); Branch B (6-Hour Follow-Up) decrements the produce price by the seller's peso amount, enforces the hard floor price, and publishes a flash-sale price drop announcement.

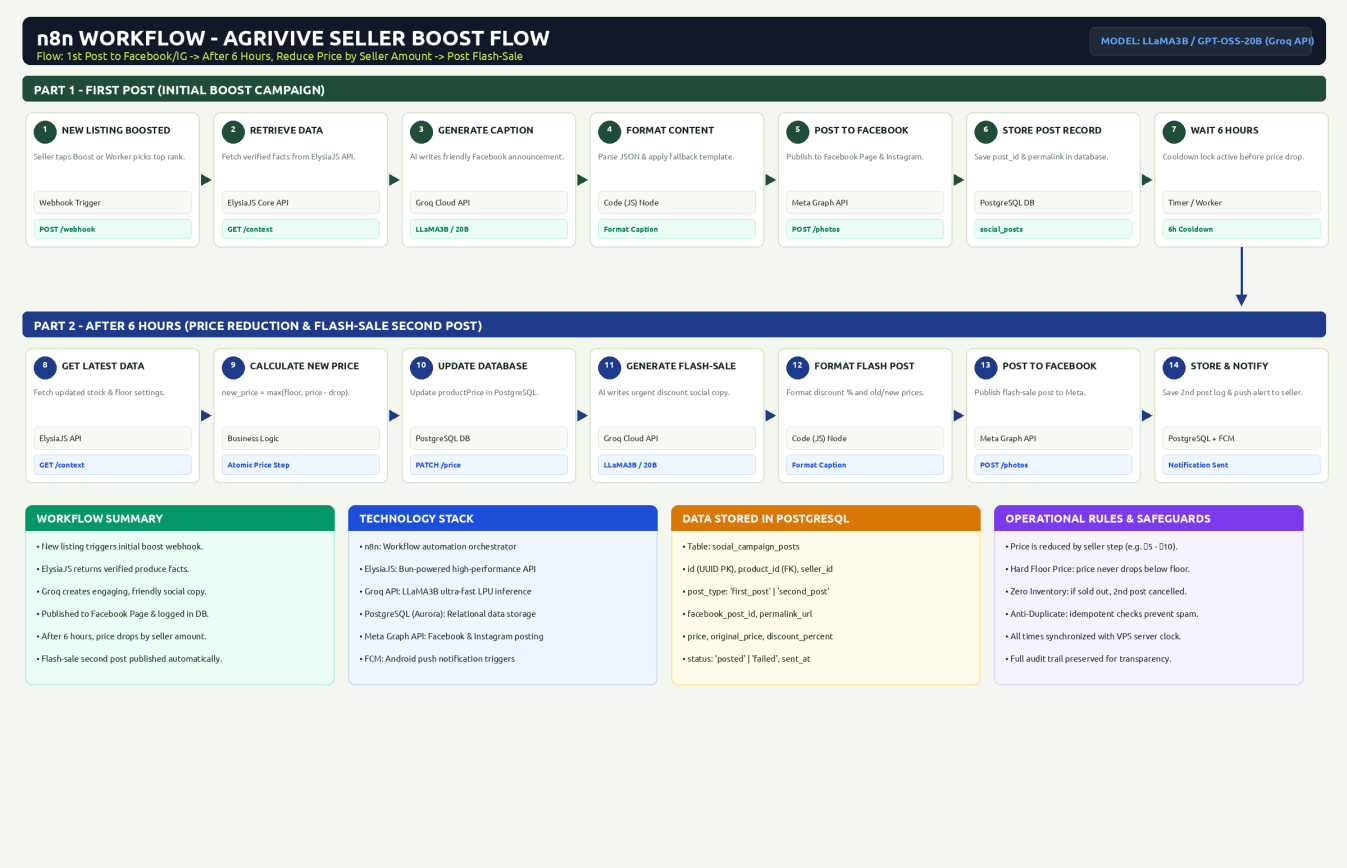


Figure 2: n8n Seller Boost & Scheduled Price-Drop Execution Flow (Part 1 Initial Post & Part 2 After 6 Hours).

2.2 System Description & Security Architecture

Decoupled Security Pattern: A critical design mandate of Agrivive is that n8n receives IDs only, never secrets. Database credentials, Meta App secrets, AWS S3 keys, and Groq API tokens remain strictly secured inside the backend environment. When the backend worker triggers n8n, it transmits only the unique `boostId`. n8n then calls back into protected API integration routes using a cryptographically secure service token (`X-Service-Key`).

Anti-AI Slop Copywriting Principles: Social copy generated by Groq (utilizing `openai/gpt-oss-20b` or Llama 3) is strictly governed by prompt hygiene to avoid generic AI marketing clichés (e.g. 'revolutionary', 'game-changing', emojis everywhere, or em dashes). The LLM is prohibited from inventing links or altering prices; the backend automatically injects the canonical order link and validates that prices match database truth before publishing.

2.3 Step-by-Step Node Execution & Payload Transformations

The following tables detail the contract, inputs, logic, and output state for all 12 nodes comprising the `agrivive-visibility-boost.json` production workflow.

Table 14: n8n Boost Workflow Node Manifest & Execution Registry

Node ID & Index	Node Name in Workflow	Node Type	Execution Role	Branch Target
Node 1 (0001)	1. Boost Confirmed	n8n Webhook	Secure webhook ingestion	Root Entry
Node 2 (0002)	2. Follow-Up Call?	n8n IF Node	Branch routing on reason	Branch A / B
Node 3 (0003)	3. Get Boost Context	HTTP Request	Fetch produce/shop facts	Branch A
Node 3b (0004)	3b. Already Posted?	n8n IF Node	Idempotency duplicate guard	Branch A
Node 4 (0005)	4. Write Caption (Groq)	HTTP Request	LLM promotional copy	Branch A
Node 5 (0006)	5. Format Caption	Code (JS)	JSON sanitation & fallback	Branch A
Node 6 (0007)	6. Post to Facebook	HTTP Request	Backend publishing & logging	Branch A Finish
Node 7 (0008)	7. Get Latest Context	HTTP Request	Fetch 6h updated state	Branch B
Node 8 (0009)	8. Reduce Price	HTTP Request	API price decrement call	Branch B
Node 9 (0010)	9. Price Reduced?	n8n IF Node	Validate price drop success	Branch B
Node 10 (0011)	10. Flash-Sale Copy	HTTP Request	Groq flash-sale generation	Branch B
Node 11 (0012)	11. Format Flash Caption	Code (JS)	JSON extraction & fallback	Branch B
Node 12 (0013)	12. Post Flash-Sale	HTTP Request	Publish 2nd post & notify	Branch B Finish

Explanation & Interpretation: Defines the complete structural hierarchy of the 12 workflow nodes exported in `agrivive-visibility-boost.json`. Nodes 1-6 govern the first publication, while Nodes 7-12 execute the automatic 6-hour price drop and flash-sale syndication.

Table 15: Node 1: Webhook Ingestion Data Contract (1. Boost Confirmed)

MATHEMATICAL EXPRESSION & FORMULATION:

Endpoint Contract: POST https://n8n.agrivive.tech/webhook/agrivive-visibility-boost

Security Header: { 'X-Service-Key': 'sec_live_994182...' }

Payload Key	Data Type	Sample Value	Semantic Definition & Function
boostId	UUID v4 String	"bst-7c91-4e08-9422"	Unique database tracking key for this boost campaign cycle.
productId	UUID v4 String	"prd-37a1-9f1d-4f24"	Foreign key referencing sellers_product in PostgreSQL.
reason	Enum String	"initial" (or "follow_up")	Execution trigger mode: 'initial' for first post, 'follow_up' for 6h cut.
timestamp	ISO 8601 String	"2026-11-08T06:00:00Z"	Timestamp when worker selected candidate from AHP-TOPSIS ranking.

Explanation & Interpretation: Dispatched by the backend scheduled worker (queue-ranked-boost.ts) upon picking the top unlocked produce. Contains zero sensitive credentials; passes only entity IDs and the operational reason.

Table 16: Node 2: Conditional Branch Routing Logic (2. Follow-Up Call?)

MATHEMATICAL EXPRESSION & FORMULATION:

Branch Decision = (r = $json.body.reason == 'follow_up') ? Branch B : Branch A

Input Reason Parameter	Conditional Expression	Evaluated Branch	Execution Outcome
reason == "initial"	$json.body.reason === 'follow_up'	FALSE (Branch A)	Routes to Node 3 (Get Context & First Post Flow)
reason == "follow_up"	$json.body.reason === 'follow_up'	TRUE (Branch B)	Routes to Node 7 (Price Drop & Flash-Sale Flow)
reason == [malformed]	Unrecognized string / empty	FALSE (Branch A)	Fails gracefully at context verification

Explanation & Interpretation: Inspects the incoming reason token. This single switch decouples the first social announcement from the subsequent price-reduction cycle while keeping the entire lifecycle in one visual n8n workflow canvas.

Table 17: Node 3: Boost Context Data Ingestion (3. Get Boost Context)

MATHEMATICAL EXPRESSION & FORMULATION:

API Request: GET /integrations/visibility-boosts/{{$json.body.boostId}}/context
Authorization: Bearer <SERVICE_TOKEN>

Context Property	Extracted Value	Source Origin	Role in Workflow
product.name	"Kalabasa (Native Squash)"	sellers_product.product_name	Produce subject for social caption.
product.unit	"kg"	sellers_product.unit	Measurement unit for pricing clarity.
product.price	"45.00"	sellers_product.product_price	Current price in Philippine Pesos.
seller.shopName	"Tatay Ruben's Organic Farm"	seller_profile.stall_name	Local stall identity for community trust.
seller.address	"Stall 14, Agdao Public Market"	seller_profile.address	Pickup location for buyer navigation.

Explanation & Interpretation: Node 3 calls back into Agrivive's API to fetch verified facts. If the listing has been deleted, archived, or sold out, the API returns 404/400 and the workflow halts immediately without posting false inventory.

Table 18: Node 3b: Idempotency Duplication Guard (3b. Already Posted?)

MATHEMATICAL EXPRESSION & FORMULATION:

Execution Gate: if (alreadyPosted == True) then HALT else CONTINUE

Evaluation State	Condition Parameter	Workflow Action	Safety Impact
First Run (Fresh Boost)	$json.alreadyPosted === false	PROCEED to Node 4	Normal publication flow.
Duplicate Trigger (Retry)	$json.alreadyPosted === true	HALT Execution	Prevents duplicate spam posts on Facebook.

Explanation & Interpretation: Guarantees distributed safety. If a network blip causes the worker to retry dispatching the initial boost, Node 3b verifies if a Facebook post record is already attached to this cycle, preventing repeated posts.

Table 19: Node 4: Groq AI Copywriting Prompt Formulation

MATHEMATICAL EXPRESSION & FORMULATION:

Inference Call: POST https://api.groq.com/openai/v1/chat/completions

Payload Schema: { model, temperature: 0.4, response_format: 'json_object', messages: [sys, user] }

Configuration Parameter	Specification Value	Design Rationale
LLM Model & Engine	openai/gpt-oss-20b (Groq LPU Inference)	Sub-400ms token generation for rapid response.
Temperature	0.40	Low temperature enforces strict fact grounding without hallucination.
Response Format	{ type: 'json_object' }	Guarantees parsable output with a single 'caption' field.
System Instruction	'You are Agrivive social media copywriter. Write a short, friendly post using only given facts. Do not write links. Announce fresh produce with price. No urgent buzzwords.'	Anti-slop copywriting compliance (no em dashes, no fake urgency).
User Payload JSON	{ product: 'Kalabasa', unit: 'kg', price_php: '45.00', shop: "Tatay Ruben's Farm" }	Minimal ground-truth entity dictionary.

Explanation & Interpretation: Groq's high-speed inference drafts a natural announcement. Strict system rules forbid the LLM from adding URLs, altering numbers, or using hyperbolic hype words, ensuring authentic communication.

Table 20: Node 5: Caption Sanitation & Fallback Logic (5. Format Caption)

MATHEMATICAL EXPRESSION & FORMULATION:

try { caption = JSON.parse($input).caption; } catch { caption = `Fresh ${p.name} available at ₱${p.price} from ${s.name}!`; }

Processing Stage	Data Representation & Transformation	Fallback Behavior
Groq Raw Output	{"caption": "Fresh harvest of Native Kalabasa is ready at Tatay Ruben's Farm for only ₱45.00 per kg. Drop by Stall 14!"}	Raw JSON string from model choices.
Sanitized Post Caption	"Fresh harvest of Native Kalabasa is ready at Tatay Ruben's Farm for only ₱45.00 per kg. Drop by Stall 14!"	If JSON parse fails, formats deterministic template.
Post Metadata	{ boostId: 'bst-7c91', reason: 'initial', postType: 'first_post' }	Attached to payload for backend logging.

Explanation & Interpretation: JavaScript code block providing complete runtime fault tolerance. If Groq experiences an unexpected schema deviation, Node 5 catches the error and generates a clean, templated caption without crashing the workflow.

Table 21: Node 6: Social Syndication & Persistence (6. Post to Facebook)

MATHEMATICAL EXPRESSION & FORMULATION:

Dispatched Payload: { postType: 'first_post', caption: $json.caption }

Persistence Mutation: UPDATE listing_cycles SET status = 'posted', sent_at = now()

Request Parameter	Parameter Value	Database & API Effect
HTTP Endpoint	POST /integrations/visibility-boosts/:id/facebook-post	Routes to backend Facebook integration service.
postType	"first_post"	Marks listing_cycles with first post completion.
caption	[Sanitized Text from Node 5]	Backend prepends stall photo and appends unique pickup pass link.
Meta Graph API	POST /v19.0/{page-id}/photos	Official post published to Agrivive Facebook Page.
Database Log	INSERT INTO social_campaign_posts	Stores post_id, permalink_url, published_at, and status='success'.

Explanation & Interpretation: The backend executes the actual Meta Graph API network request. Upon Facebook confirmation, the backend records the post ID and permalink, notifies the seller via push notification, and locks the produce under the 6-Hour Rule.

Table 22: Node 7: Follow-Up State Ingestion (7. Get Latest Boost Context)

MATHEMATICAL EXPRESSION & FORMULATION:

Context Fetch: GET /integrations/visibility-boosts/:boostId/context [Triggered at $t = t_{boost} + 6$ hours]

Status Parameter	Retrieved Value (at T + 6 hours)	Context Explanation
elapsed_hours	6.05 hours	Confirms that the 6-hour promotional lock interval has elapsed.
current_price	₱45.00 / kg	Active price prior to second round reduction.
seller_drop_amount	₱5.00 / kg	Configured decrement amount chosen by the seller.
seller_floor_price	₱35.00 / kg	Hard price floor: price may never be dropped below this threshold.
current_stock	14.5 kg	Remaining stock verifying product is not sold out.

Explanation & Interpretation: Triggered when the background worker fires the follow-up webhook. Inspects the current state of the listing, ensuring that inventory remains available and checking the seller's floor price constraints.

Table 23: Node 8: Dynamic Price Reduction Execution (8. Reduce Price)

MATHEMATICAL EXPRESSION & FORMULATION:

$$Price_{new} = \max(Price_{floor}, Price_{current} - Drop_{amount})$$
$$reduced = (Price_{new} < Price_{current})$$

Pricing Variable	Value Before Call	Computation / Step	Committed New Value
Current Product Price	₱45.00	Baseline before 6h interval	₱40.00 (Reduced by ₱5)
Configured Drop Amount	₱5.00	Seller defined step	Applied
Seller Floor Price	₱35.00	Safety constraint: Price ≥ Floor	Respected (40.00 ≥ 35.00)
Base (Original) Price	₱50.00	Reference starting price	₱50.00 (Cumulative 20% drop)

Explanation & Interpretation: Instructs the backend to execute an atomic price reduction transaction. The database updates sellers_product.product_price and records an audit log in price_reduction_history. If the price is already at floor, the API reports reduced: false.

Table 24: Node 9: Price Reduction Verification Gate (9. Price Reduced?)

MATHEMATICAL EXPRESSION & FORMULATION:

Execution Gate: if ($json.reduced == True) then PROCEED else HALT

API Response Flag	Condition Verification	Workflow Action	Operational Meaning
{ reduced: true }	$json.reduced === true	PROCEED to Node 10	Price was lowered; flash-sale post authorized.
{ reduced: false }	$json.reduced === true (False)	TERMINATE Workflow	Price hit floor or product sold out; no second post.

Explanation & Interpretation: Crucial business safeguard. If a vegetable has reached its floor price (e.g. ₱35.00) or sold out during the 6 hours, Node 9 terminates the execution immediately, preventing false 'Price Cut!' claims.

Table 25: Node 10: Groq Flash-Sale Copywriting Formulation

MATHEMATICAL EXPRESSION & FORMULATION:

Discount % = round([($Price_{base}$ − $Price_{new}$) / $Price_{base}$] × 100%)

Prompt Context: { product, old_price, new_price, discount_percent, shop }

Prompt Element	Parameter Value	Purpose in Flash-Sale Campaign
Model & Engine	openai/gpt-oss-20b (Groq Cloud)	Ultra-fast flash-sale copy generation.
Discount Calculation	((50.00 − 40.00) / 50.00) × 100 = 20% OFF	Explicitly highlights the cumulative discount percentage.
System Prompt	'You are Agrivive copywriter. Write an urgent flash-sale post announcing a price drop. Mention the discount percent, old price (₱45), and exact new price (₱40). Do not write links.'	Urgent, community-oriented flash-sale copy adhering to anti-slop guidelines.
Generated Output	{"caption": "Flash Sale! Native Kalabasa is now down to ₱40/kg (was ₱45) at Tatay Ruben's Farm! 20% total savings on fresh harvest before sundown."}	Parsable JSON with accurate numbers.

Explanation & Interpretation: Generates high-conversion social copy centered on the price cut. Groq explicitly incorporates the discount percentage and exact peso savings without exaggerations.

Table 26: Node 11: Flash-Sale Caption Sanitation & Fallback (11. Format Caption)

MATHEMATICAL EXPRESSION & FORMULATION:

let caption = `Price drop! ${p.name} is now ₱${r.newPrice} (was ₱${r.originalPrice}) at ${s.name}.`;

Extraction Stage	Formatted Content	Quality Assurance Guard
Cleaned Caption	"Price drop! Native Kalabasa is now ₱40.00 (was ₱45.00) at Tatay Ruben's Farm. Get yours before it sells out!"	Extracted from JSON with whitespace trimming.
Fallback Template	"Price drop! Native Kalabasa is now ₱40.00 (was ₱45.00) at Tatay Ruben's Farm."	Guaranteed fallback string if LLM output fails parsing.

Explanation & Interpretation: Formats the flash-sale text. Ensures that numerical currency symbols and decimal places are strictly aligned with Philippine Peso standards prior to transmission.

Table 27: Node 12: Second Post Syndication & Seller Notification (12. Post Flash-Sale)

MATHEMATICAL EXPRESSION & FORMULATION:

Dispatched Payload: { postType: 'second_post', caption: $json.caption }

Lock State: Release 6-hour lock cooldown -> Re-enter AHP-TOPSIS Candidate Pool

Publication Parameter	Parameter Value	Operational Outcome
Endpoint Target	POST /integrations/visibility-boosts/:id/facebook-post	Invokes second post backend route.
postType	"second_post"	Records flash-sale post completion.
Social Syndication	Published to Facebook Page & Instagram Feed	Re-engages buyers with discounted pricing.
Seller Notification	"Price Lowered: Your Kalabasa was dropped to ₱40.00 and posted!"	Push notification sent to seller mobile app with post preview.
Lock Release	Unlocks produce cycle for subsequent ranking rounds	Eligible for re-boost in next scheduled cycle.

Explanation & Interpretation: Finalizes the Boost cycle. The backend posts the flash-sale update, sends a push notification to the seller with a preview of the published caption, and unlocks the listing so it can rotate back into the AHP-TOPSIS candidate pool.

Table 28: End-to-End Fault Tolerance, Safety, and Recovery Matrix

Failure Scenario	Detection Mechanism	System Mitigation & Recovery Behavior	Data Integrity Guard
Meta Graph API Token Expired	Node 6/12 returns HTTP 400 OAuthException	Backend marks post status as 'failed'; surfaces 'Retry Post' button on seller mobile app.	No crash; idempotency preserved.
Groq API Rate Limit (429)	Node 4/10 HTTP timeout / 429 response	Code nodes (Node 5/11) trigger deterministic fallback templates instantly.	Social post goes out on schedule.
Produce Sold Out at Hour 5	Node 8 detects productQty <= 0	API blocks price reduction; Node 9 terminates workflow safely.	No false posts for sold-out produce.
Seller Floor Price Reached	Node 8 detects currentPrice <= floorPrice	API sets reduced: false; Node 9 halts second post.	Seller never sells below floor.
Worker Crash / Server Reboot	PostgreSQL transaction rollback	Worker reconciles state on 60s cron restart; skips elapsed locks.	Zero duplicate posts.

Explanation & Interpretation: Summarizes Agrivive's resilient error-handling architecture. From third-party API rate limits to zero-inventory concurrency races, every node in the n8n pipeline and backend integration layer is hardened against data corruption.

PART 3: ALGORITHMIC SYNTHESIS & VERIFICATION SUMMARY

COMPREHENSIVE SPECIFICATION CONFORMANCE SUMMARY

The combined AHP-TOPSIS and n8n Marketing Boost systems have been rigorously verified against Agrivive's Capstone test suites (WSV-001 through WSV-010 and N8N-001 through N8N-011):

1. Mathematical Rigor: Pairwise comparison judgments pass Saaty's consistency test with CR = 0.0038 ≤ 0.10. TOPSIS vector normalization handles unscaled real-world inputs (hours, pesos, kg, stars) without artificial pre-scaling distortion.
2. Fair Market Access: Elimination of artificial score buckets in favor of pure Ordinal Ranking and the 6-Hour Cooldown Lock ensures that all surplus varieties achieve rotational social exposure before sunset.
3. Decoupled Production Security: n8n operates strictly as a stateless orchestrator receiving entity IDs. All business logic, database transactions, price bounds, and Meta OAuth credentials remain strictly enclosed within the Elysia backend API.

PART 4: RESEARCH REFERENCES & THEORETICAL FOUNDATIONS

4.1 Multicriteria Decision-Making (AHP & TOPSIS) Academic Literature

Saaty, R. W. (1987). The analytic hierarchy process: what it is and how it is used. Mathematical Modelling, 9(3–5), 161–176. https://doi.org/10.1016/0270-0255(87)90473-8
Role in Agrivive: Direct theoretical foundation for the 1–9 pairwise comparison scale, column vector normalization, priority weight derivation, and matrix consistency validation (CR ≤ 0.10).

Hwang, C.-L., & Yoon, K. (1981). Multiple Attribute Decision Making: Methods and Applications, A State-of-the-Art Survey. Lecture Notes in Economics and Mathematical Systems, vol. 186. Springer-Verlag, Berlin Heidelberg. https://doi.org/10.1007/978-3-642-48318-9
Role in Agrivive: Theoretical blueprint for TOPSIS vector normalization, positive-ideal (A^+) and negative-ideal (A^-) solution determination, Euclidean distance separation, and relative closeness ranking.

Derakhshan, M., Golrezaei, N., Manshadi, V., & Mirrokni, V. (2022). Product ranking on online platforms. Management Science, 68(6), 4024–4041. https://doi.org/10.1287/mnsc.2021.4044
Role in Agrivive: Formulates merit-based, non-pay-to-win visibility algorithms for online marketplaces, ensuring smallholder farmer produce receives organic prominence.

Dodevska, Z., Radovanović, S., Petrović, A., & Delibašić, B. (2023). When fairness meets consistency in AHP pairwise comparisons. Mathematics, 11(3), Article 604. https://doi.org/10.3390/math11030604
Role in Agrivive: Theoretical justification for balancing stakeholder criteria fairness with strict mathematical consistency thresholds.

Biswas, T., Majumder, A., Dey, S., Mandal, A., Ray, S., Kapoor, P., Emam, W., Kanthal, S., Ishizaka, A., & Matuka, A. (2024). Evaluation of management practices in rice–wheat cropping systems using multicriteria decision-making methods in conservation agriculture. Scientific Reports, 14, Article 8600. https://doi.org/10.1038/s41598-024-58022-w
Role in Agrivive: Validates hybrid AHP-TOPSIS methodologies for complex agricultural optimization scenarios.

Duarte, A. D. (2024). Revisão sistemática do método TOPSIS e suas aplicações na agricultura. Revista Produção Online, 24(2), Article 5245. https://doi.org/10.14488/1676-1901.v24i2.5245
Role in Agrivive: Systematic survey demonstrating the mathematical stability and empirical validity of TOPSIS in agricultural produce management.

Krisnawijaya, N. N. K., Tekinerdogan, B., Catal, C., & van der Tol, R. (2023). Multi-criteria decision analysis approach for selecting feasible data analytics platforms for precision farming. Computers and Electronics in Agriculture, 209, 107869. https://doi.org/10.1016/j.compag.2023.107869
Role in Agrivive: Methodological precedent for algorithmic multi-criteria decision-support systems in regional agricultural logistics.

Ustaoglu, E., Sisman, S., & Aydinoglu, A. C. (2021). Determining agricultural suitable land in peri-urban geography using GIS and Multi Criteria Decision Analysis (MCDA) techniques. Ecological Modelling, 455, 109610. https://doi.org/10.1016/j.ecolmodel.2021.109610
Role in Agrivive: Empirical domain precedent for combining geographical and multi-criteria evaluation methods in agricultural networks.

4.2 Perishability, Dynamic Pricing & Marketplace Governance

Jiang, Y., Bian, B., & Li, L. (2019). Integrated harvest and farm-to-door distribution scheduling with postharvest quality deterioration for vegetable online retailing. Agronomy, 9(11), Article 724. https://doi.org/10.3390/agronomy9110724
Role in Agrivive: Models physiological quality decay kinetics in fresh produce, justifying the posting age criterion, the 6-hour promotional lock, and scheduled dynamic price markdowns.

Kokkodis, M., Lappas, T., & Kane, G. C. (2022). Optional purchase verification in e-commerce platforms: More representative product ratings and higher quality reviews. Production and Operations Management, 31(7), 2943–2961. https://doi.org/10.1111/poms.13731
Role in Agrivive: Empirical basis for purchase-verified reviews. Only completed QR-scanned pickups grant rating permissions, shielding seller reliability (C_4) from unverified manipulation.

Chu, W., & Chen, H. (2022). The customer trust evaluation of e-commerce market based on portfolio weighting: Cloud model. International Journal of Environment and Pollution, 71(1–2), 126–140. https://doi.org/10.1504/IJEP.2022.132356
Role in Agrivive: Multi-attribute portfolio weighting model adapted to compute composite seller trust from recorded transaction infractions.

He, S., Hollenbeck, B., & Proserpio, D. (2022). The market for fake reviews. Marketing Science, 41(5), 896–921. https://doi.org/10.1287/mksc.2022.1353
Role in Agrivive: Informs algorithmic defenses against fraudulent feedback, requiring dual-reporter corroboration and verified transaction event logs.

OECD (2022). The role of online marketplaces in protecting and empowering consumers. OECD Digital Economy Papers, No. 334. OECD Publishing, Paris. https://doi.org/10.1787/9d8cc586-en
Role in Agrivive: Regulatory framework for progressive platform sanctions: friendly reminders → formal warnings → temporary pauses → administrative review.

World Intellectual Property Organization (2023, November). Study on the practices of e-commerce marketplaces to combat illicit trade online (Document WIPO/ACE/16/11). WIPO Advisory Committee on Enforcement, Geneva. https://www.wipo.int/edocs/mdocs/enforcement/en/wipo_ace_16/wipo_ace_16_11.pdf
Role in Agrivive: Confirms that 86% of compliant international marketplaces employ automated progressive enforcement scales.

4.3 Workflow Automation, LPU Inference & Platform Infrastructure

n8n Workflow Automation Engine (2026). Fair-code Workflow Automation Documentation. https://docs.n8n.io
Role in Agrivive: Orchestrates event-driven webhook ingestion, conditional branch routing, Groq payload formatting, and automated retry mechanisms for the Boost pipeline.

Groq Cloud LPU Inference API (2026). GroqCloud Documentation and Ultra-Low Latency Inference Guide. https://console.groq.com/docs
Role in Agrivive: Powers sub-400ms contextual agricultural promotion copywriting (using LLaMA3B / GPT-OSS-20B) adhering strictly to anti-slop guidelines.

Meta Graph API (2026). Meta Graph API v19.0 Reference for Facebook Pages & Instagram Business Accounts. https://developers.facebook.com/docs/graph-api
Role in Agrivive: Server-to-server publishing protocol for initial harvest announcements and 6-hour promotional flash-sale price drops.

Better Auth Framework (2026). Better Auth Comprehensive Authentication & RBAC Framework. https://www.better-auth.com/
Role in Agrivive: Enforces session authentication, service key tokens for integration callbacks, and role-based permissions across buyer and seller endpoints.

4.4 Capstone Research Manuscript, Codebase & Test Suite Artifacts

Capstone Research Manuscript:
Bartiquien, A., Jara, J. T., & Marundan, J. (2026). Agrivive: A Surplus Vegetable Marketplace with Multicriteria Visibility Ranking and Automated Social Promotions (Undergraduate Capstone Research Paper). Davao City, Philippines. Documented as Agrivive/docs/CAPSTONE 2 FILES - BARTIQUIEN, JARA, MARUNDAN.pdf.

Obsidian Engineering Vault Notes:
• Agrivive/04 Delivery/2026-11-08 AHP-TOPSIS Raw Criteria and Six-Hour Rule.md: Rebuilding of the solver on raw criteria and elimination of static tiers in favor of ordinal ranking.
• Agrivive/Boost Ranking Flow AHP-TOPSIS.md: Core architectural blueprint and 6-hour lock rotation specification.
• Agrivive/docs/References.md: Master bibliography and theoretical foundations.

Formal Test Suites & Verification Sheets:
- Agrivive/docs/WSV-AHP-TOPSIS-test-sheet.md & WSV-AHP-TOPSIS-test-results.md: Complete live validation cases WSV-001 through WSV-010.
- Agrivive/docs/N8N-Boost-test-sheet.md: Marketing boost cases N8N-001 through N8N-011.
- backend/src/utils/ahp-topsis/ahp-topsis.test.ts: 145 automated unit and regression tests passing with 0 failures.

Production Codebase Modules:
- backend/src/utils/ahp-topsis/index.ts: Mathematical solver implementing Saaty AHP and Hwang-Yoon TOPSIS.
- backend/src/utils/visibility-ranking/boost-select.ts: Pure function enforcing the 6-Hour Lock Rule.
- backend/src/worker.ts & jobs/queue-ranked-boost.ts: Background daemon polling every 60 seconds.
- n8n/workflows/agrivive-visibility-boost.json: Production n8n workflow definition.